MediVault

Advanced Health Records & Emergency Access System

Technical Architecture & Implementation Guide

1. System Overview

MediVault is a comprehensive medical records management platform designed to bridge the gap between patient data privacy and emergency accessibility. It provides two distinct yet interconnected layers: a highly secure **Patient-Doctor ecosystem** and a rapid-response **Emergency Mode**.

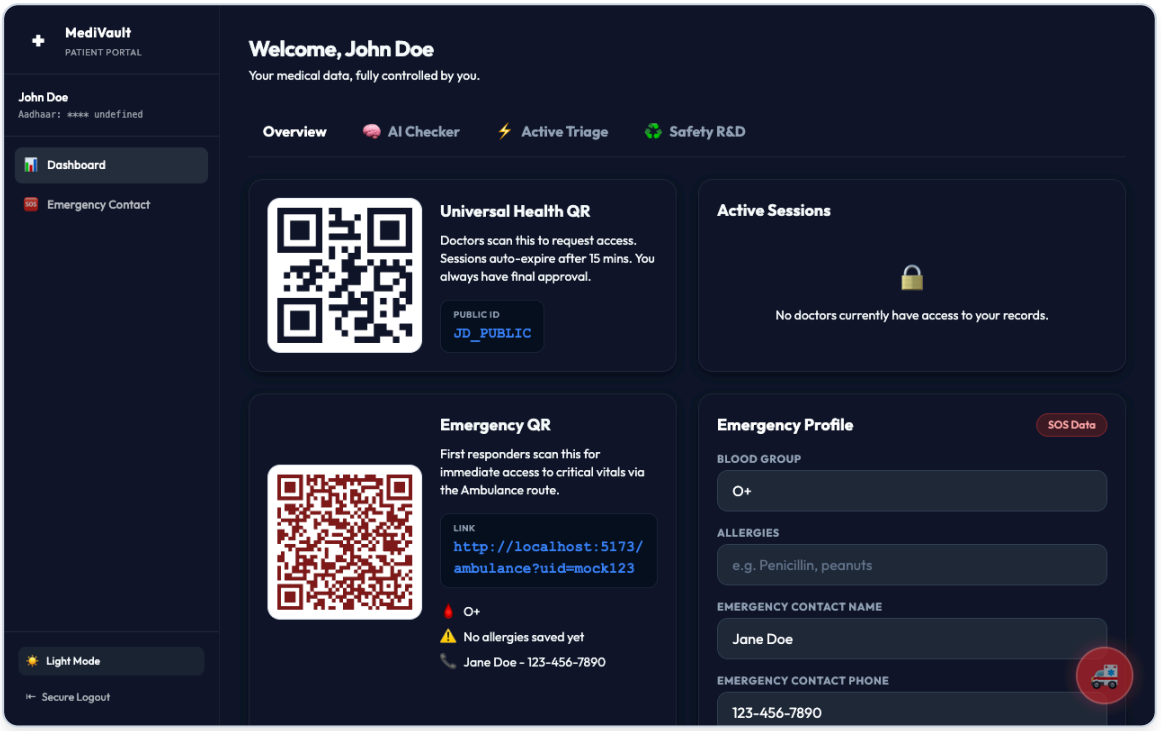


Figure 1.1: The main Patient Dashboard showing Medical History

2. Technology Stack

Frontend Architecture

- **React 18** UI Library
- **Vite** Build Tool & Bundler
- **React Router** Client Navigation

Backend & Infrastructure

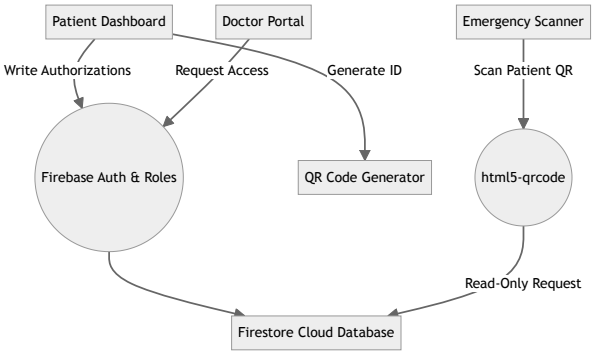
- **Firebase Auth** Security & Roles
- **Firestore DB** Realtime NoSQL Database
- **Service Workers** Offline Capabilities

Tailwind CSS Styling Engine

Framer Motion Animations & Transitions

3. Architectural Flowcharts

Below is the core Data Architecture Flow.



4. Core Module Implementations

4.1 Emergency Mode

Designed specifically for high-stress scenarios. It bypasses the standard login flow, relying strictly on cryptographically generated patient Emergency IDs encoded in QR format. The interface explicitly utilizes a dark, low-glare aesthetic to ensure optimal readability for first responders in intense situations.

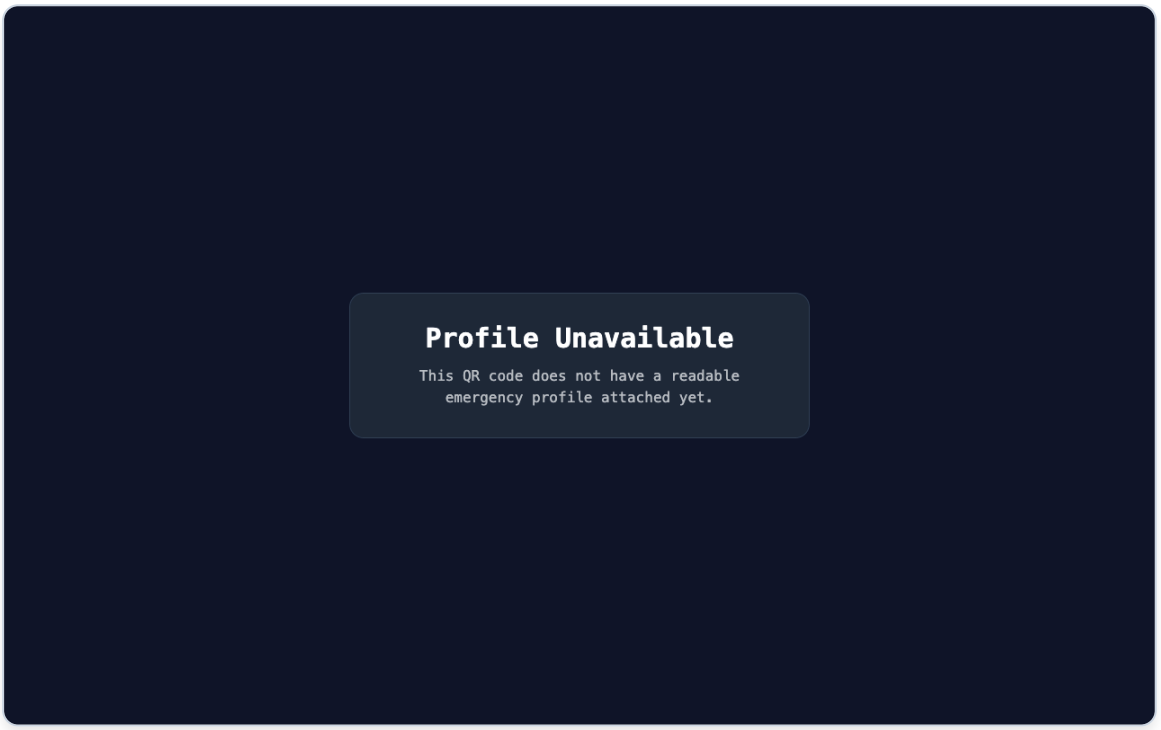


Figure 4.1: The fast-access Emergency Paramedic Dashboard